Sparse & Explainable Bayesian Neural Networks for Official Statistics

Unit-Weighted Pseudo-Posteriors, Formal Differential Privacy, and Non-Probability Propensity Estimation in LBBNN

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Section 1

PART I: Foundations of LBBNN & Input-Skip Architecture (ISLaB)

Motivation — Why Sparse & Explainable BNNs in Official Statistics?

The Black-Box & Over-Parameterization Dilemma:

- Modern deep neural networks capture complex multi-way interactions, but lack calibrated uncertainty quantification and interpretable feature attributions.
- Standard dense BNNs quantify parameter uncertainty, but retain all noise covariates, inflating variance in downstream survey estimators.

What Official Statistics Requires:

- **Structural Sparsity**: Combine model selection (spike-and-slab edge pruning) and parameter uncertainty in a single variational framework.
- **Explainable Depth**: Automatically adapt between simple linear GLMs (Depth $= 1$) and deep non-linear interaction surfaces (Depth > 1).
- **Survey & Privacy Rigor**: Rigorous integration with informative sampling weights, differential privacy (ϵ, δ), and non-probability samples.

From Dense BNNs to Latent Binary BNNs (Hubin & Storvik, 2024)

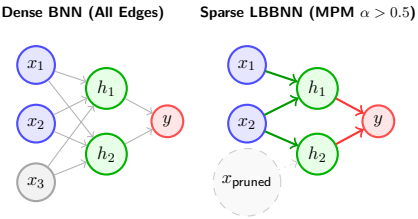
Spike-and-Slab Edge Selection:

- Latent binary inclusion indicator $\gamma_{k,p}^{(j)} \in \{0, 1\}$:

$$p\big(w_{k,p}^{(j)} \mid \gamma_{k,p}^{(j)}\big) = \gamma_{k,p}^{(j)} \mathcal{N}\big(0, (\tau_{k,p}^{(j)})^2\big) + \big(1 - \gamma_{k,p}^{(j)}\big)\delta_0$$

$$p\big(\gamma_{k,p}^{(j)}\big) = \text{Bernoulli}\big(\psi_{k,p}^{(j)}\big)$$

- **Bayesian Model Averaging (BMA)**: Ensembles across posterior draws $\Gamma \sim q(\Gamma)$ and $W \sim q(W \mid \Gamma)$ for optimal uncertainty quantification.
- **Median Probability Model (MPM)**: Hard-prunes edges with inclusion probability $\alpha_{k,p}^{(j)} \leq 0.5$ (Barbieri & Berger, 2004 optimal predictive submodel).



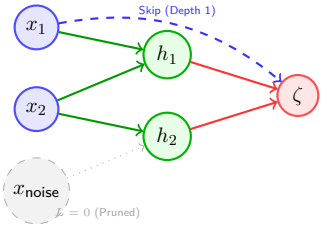
Input-Skip LBBNNs (ISLaB) & Structural Depth Metrics (Høyheim et al., 2026)

The ISLaB Architecture (*JAIR*, 2026):

- Direct skips route inputs $\mathbf{x} \in \mathbb{R}^p$ to every hidden layer: $\mathbf{a}^{(j-1)} = [a_1^{(j-1)}, \ldots, a_K^{(j-1)}, \mathbf{x}^\top]^\top$.
- **Active Path (AP)**: Connects input x_k to output ζ via edges with $\alpha_{k,p}^{(j)} > 0.5$.

Contribution Depth Classification:

- **Depth = 1**: Direct skip $\implies$ linear GLM effect.
- **Depth $\geq$ 2**: Hidden layers $\implies$ non-linear interaction.
- **Paths = 0**: Noise covariate pruned from graph.



Local Explainability via Polyhedral ReLU Activation Regions

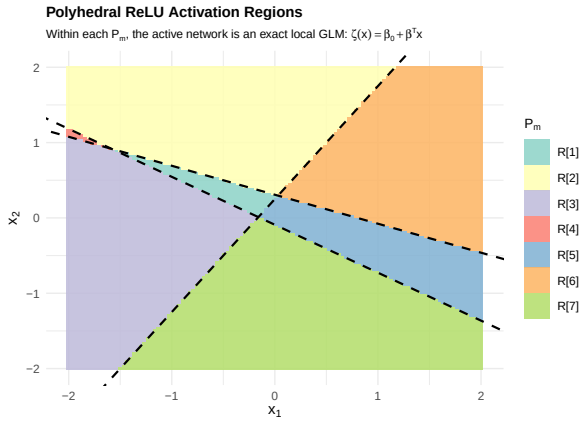
Local GLM Theorem (JSM 2026):

- Piecewise-linear activations partition $\mathbb{R}^p$ into polyhedral activation regions $\mathcal{P}_1, \ldots, \mathcal{P}_M$.
- Within each local region $\mathcal{P}_m$, the active network reduces to an exact **local GLM**:

$$\zeta(\mathbf{x}) = \beta_0(\mathbf{x}_i) + \sum_{j=1}^{p} \beta_j(\mathbf{x}_i) x_j$$

$$\boldsymbol{\beta}(\mathbf{x}_i) = \nabla_{\mathbf{x}} \zeta(\mathbf{x})\big|_{\mathbf{x}=\mathbf{x}_i}$$

- **Exact Gradients**: Evaluated via analytical backpropagation (zero perturbation error).
- **Credible Intervals**: Draws $W \sim q(W \mid \Gamma^*)$ yield exact posterior intervals for respondent sensitivities.



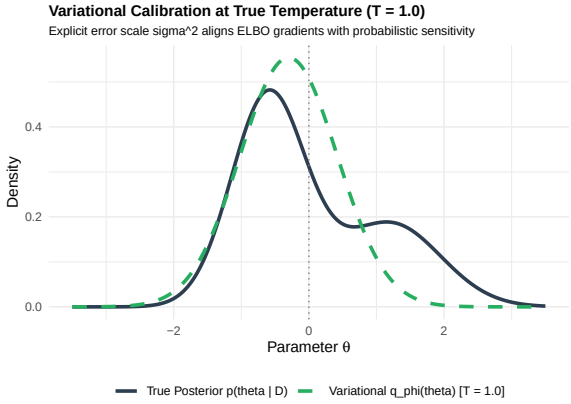
Variational Inference & Protection Against Cold Posteriors

Variational ELBO at True $T = 1.0$:

$$\text{ELBO}(q) = \sum_{i=1}^{n} \mathbb{E}_q[\log f(y_i \mid \mathbf{x}_i, \theta)] - \text{KL}(q \parallel p)$$

Why Explicit Error Scale σ^2 Matters (Wenzel et al., 2020):

- Omitting residual noise scale σ^2 artificially rescales likelihood ($T = \sigma^2 \neq 1$), mis-calibrating spike-and-slab KL shrinkage.
- In `lbbnn_net(sigma = ...)`, calibrated σ^2 grounds log-likelihood sensitivities on the true Bayesian scale.
- Calibrates both feature inclusion probabilities α_k and DP sensitivities f_i.



The LBBNN Open-Source Software Ecosystem

R Package LBBNN (CRAN v0.1.6):

- Pure C++ LibTorch backend via `torch` for R (zero Python dependencies or overhead).
- Mean-field spike-and-slab with Local Reparameterization Trick (LRT) and Normalizing Flows (*TMLR*, 2024).
- Automated model selection, predictive ensembling (BMA), and sparse median probability models (MPM).

Joint BLS Collaboration Modules:

- **Unit Weighting**: `get_dataloaders()` supports non-trainable weights $u_i \geq 0$ with optional sample-size normalization.
- **Residual Calibration**: `lbbnn_net(sigma = ...)` incorporates observation noise scale.
- **Differential Privacy**: `loglik_contributions()` and `fit_dp_lbbnn()` for two-pass (ϵ, δ)-PDP synthesis.
- **Composite Propensities**: Native losses for `STACK`, `LINK1`, `LINK2`, and `MIP`.

Section 2

PART II: Track 1 — Unit-Weighted Pseudo-Posteriors & Formal Differential Privacy

Unit-Weighted Variational Pseudo-Posteriors (Savitsky & Toth, 2016)

Equipping each unit $i \in \{1, \ldots, n\}$ with a non-negative weight $u_i \geq 0$ defines the pseudo-posterior:

$$p_{\mathbf{u}}(\theta \mid \mathbf{y}) \propto p(\theta) \prod_{i=1}^{n} f(y_i \mid \mathbf{x}_i, \theta)^{u_i}$$

$$\mathsf{ELBO}(q; \mathbf{u}) = \sum_{i=1}^{n} u_i \, \mathbb{E}_{q(\theta)}[\log f(y_i \mid \mathbf{x}_i, \theta)] - \mathsf{KL}\big(q(\theta) \parallel p(\theta)\big)$$

1. Survey Sampling Weights ($u_i = \tilde{w}_i$):

- Normalized to sample size n: $\tilde{w}_i = w_i \frac{n}{\sum w_j}$.
- Setting `normalize_weights = TRUE` ensures $\sum \tilde{w}_i = n$ does not overpower spike-and-slab KL shrinkage.
- Corrects informative sampling selection bias.

2. Differential Privacy Weights ($u_i = \alpha_i$):

- Unnormalized weights: $\alpha_i \in [0, 1]$.
- Setting `normalize_weights = FALSE` ensures $\sum \alpha_i \leq n$.
- Downweights high-sensitivity tail outliers and contracts the pseudo-posterior toward the prior.

Track 1A — Informative Survey Sampling Setup & Automated Noise Pruning

Simulation Setup ($N = 2,000, p = 4$):

- **True Non-Linear DGP**:

$$Y_i = 2X_{1i} + 1.5\sin(\pi X_{2i}) + 0.8X_{1i}X_{2i} + \varepsilon_i$$

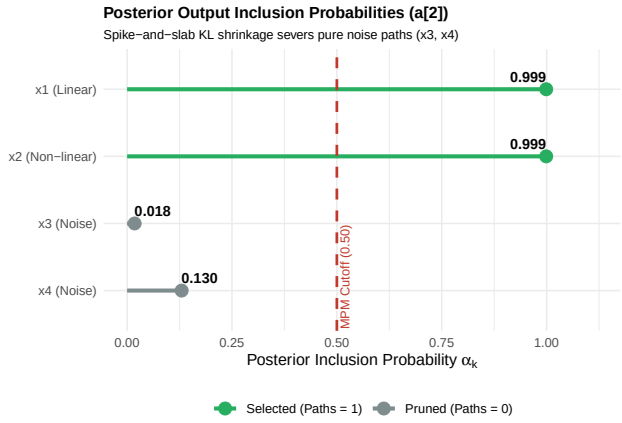
where $\varepsilon_i \sim \mathcal{N}(0, 0.4^2)$, and $X_3, X_4 \sim \mathcal{N}(0, 1)$ are pure noise.

- **Informative Selection Mechanism**:

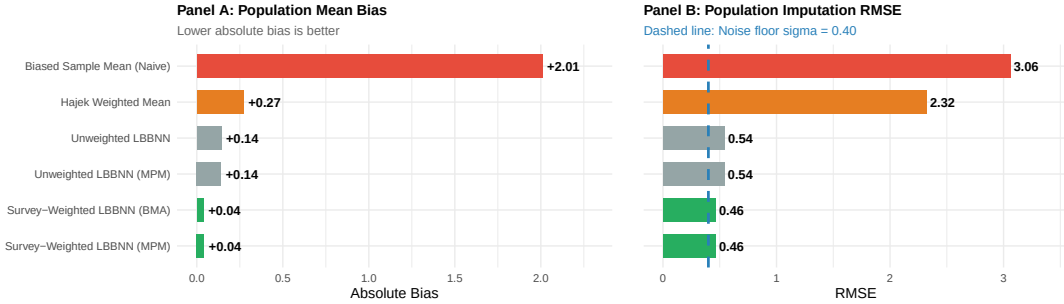
$$\pi_i = \text{logit}^{-1}(-1.0 + 1.2Y_i)$$

Draws $n = 737$ units favoring high Y_i.

- Unweighted sample mean is **+2.009** (True population mean: **0.000**).



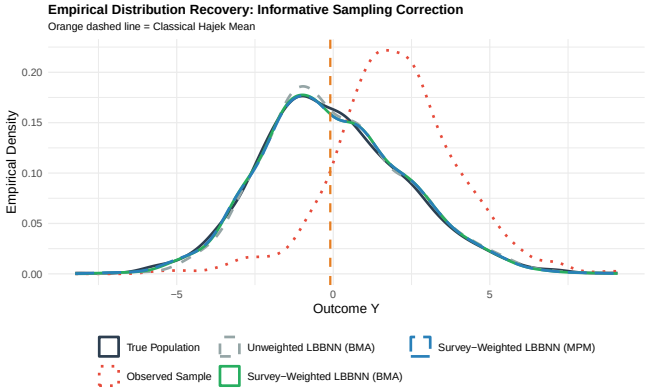
Track 1A — Population Mean & Imputation RMSE Recovery (Visual Benchmark)



Key Takeaways:

- Classical **Hájek mean** reduces bias ($+0.27$) but only provides a single scalar estimate (Pop RMSE $= 2.32$).
- **Survey-Weighted LBBNN** learns the conditional surface, reducing Pop RMSE to **0.46** (approaching $\sigma = 0.40$ noise floor).

Track 1A — Full Population Distribution Recovery



Distributional Recovery: Survey-weighted LBBNN models (green/blue curves) accurately reconstruct the full population density (black solid curve) across all 2,000 units, completely correcting the severe rightward shift of the observed sample (red dotted curve).

Track 1B — Why Unweighted Neural Networks Fail Differential Privacy

Probabilistic Differential Privacy (Savitsky, Williams & Hu, 2022):

$$\sup_{\theta \in \Theta_\delta} \left| \log \frac{q_{\boldsymbol{\alpha}}(\theta \mid \mathbf{y})}{q_{\boldsymbol{\alpha}}(\theta \mid \mathbf{y}')} \right| \leq \epsilon_{\mathbf{y}} = 2\Delta_{\alpha,\mathbf{y}}, \qquad \Delta_{\alpha,\mathbf{y}} = \max_{s,i} |\alpha_i \log f(y_i \mid \mathbf{x}_i, \theta^{(s)})|$$

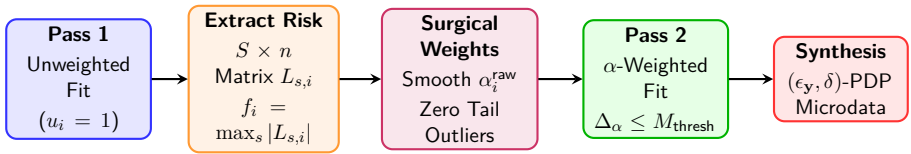
The Quadratic Exposure Dilemma in Gaussian Regression:

- Pointwise Gaussian log-likelihood grows quadratically with the residual:

$$|\log f(y_i \mid \mathbf{x}_i, \theta)| \propto \frac{(y_i - \mu_\theta(\mathbf{x}_i))^2}{2\sigma^2}$$

- In skewed survey microdata (wages, financial expenditures), extreme outliers drive unweighted sensitivity $\Delta \to \infty$ and privacy budget $\epsilon_{\mathbf{y}} \to \infty$.
- Naive gradient clipping distorts the mean and destroys the variance structure of official statistics.

Track 1B — The Two-Pass Surgical Downweighting Pipeline (`fit_dp_lbbnn`)

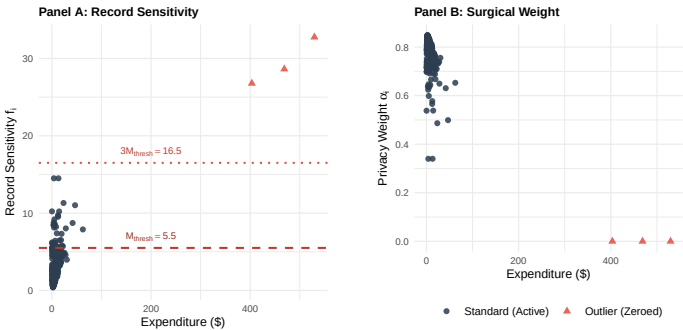


Monotone Surgical Truncation Formula:

$$\alpha_i = \min\left(\alpha_i^{\text{raw}}, \frac{M_{\text{thresh}}}{f_i}\right), \qquad \alpha_i = 0 \quad \text{if } f_i > 3M_{\text{thresh}}$$

Guarantees local Lipschitz bound $\Delta_{\alpha,\mathbf{y}} = \max_{s,i} |\alpha_i L_{s,i}^{(2)}| \leq M_{\text{thresh}}$ and bounded privacy $\epsilon_{\mathbf{y}} = 2\Delta_{\alpha,\mathbf{y}}$.

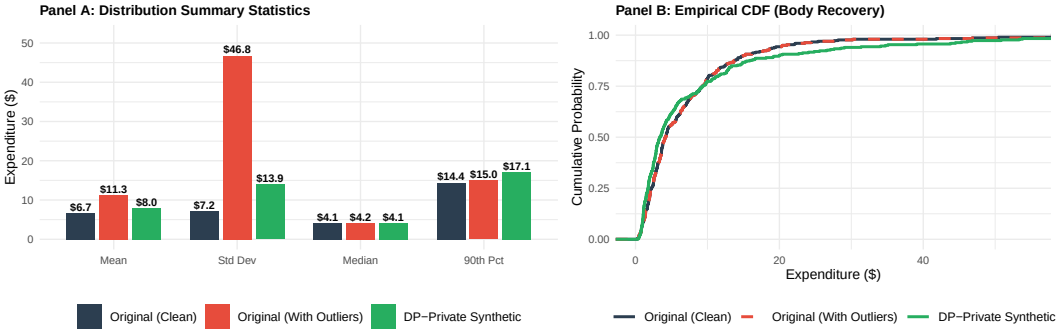
Track 1B — Monotone Surgical Downweighting Diagnostic



Empirical DP Metrics:

- **Sample**: $n = 300$, median \$4.16, 3 extreme outliers (\$400–\$530).
- **Standard Records**: Retain high weights $\alpha_i \in [0.34, 0.85]$ (median 0.81).
- **Tail Outliers**: All 3 outliers $(f_i > 26 > 3M)$ surgically zeroed $(\alpha_i = 0.0)$.
- **Privacy Guarantee**: $\Delta_{\alpha,\mathbf{y}} = 4.32 \leq 5.50 \implies \boldsymbol{\epsilon_{\mathbf{y}} = 8.63}$.

Track 1B — Synthetic Microdata Utility vs. Outlier Protection



Distributional Utility vs. Disclosure Protection:

- 3 extreme outliers distort raw sample mean ($\$11.30$ vs. $\$6.70$) and explode standard deviation to $\$46.75$.
- **DP-Private Synthetic** microdata recovers the true distribution body (Mean $\$7.99$, Median $\$4.13$) while safely truncating disclosure risk on extreme tail outliers.

Section 3

PART III: Track 2 — Combining Probability & Non-Probability Samples

The Big Data Paradox in Convenience Samples (Meng, 2018)

The Non-Probability Selection Dilemma:

- In convenience/opt-in samples S_c, sample inclusion $I_{ci} \in \{0, 1\}$ is non-random and driven by unobserved self-selection mechanisms.
- **Meng's Law of Large Populations (2018)**:

$$\text{Bias}(\bar{y}_c) = \frac{\text{Cov}_U(I_c, Y)}{\bar{\pi}_c} = \rho_{I_c,Y} \cdot \sigma_Y \cdot \sqrt{\frac{N - n_c}{n_c}}$$

- A tiny correlation $\rho_{I_c,Y} \neq 0$ creates selection bias that scales with $\sqrt{N}$.
- Merely increasing convenience sample size n_c shrinks variance around a severely biased estimate (the Big Data Paradox).

The Identification Barrier & The Conditioning Trick (Beresovsky et al., 2026)

The Identification Barrier:

- We observe covariates $\mathbf{x}_i$ only on convenience sample S_c (unknown π_{ci}) and reference survey S_r (known design probability π_{ri}), never on $U \setminus (S_c \cup S_r)$.

The Conditioning Trick:

- Pool S_c and S_r into a combined sample S, and model membership indicator $D_i = \mathbb{I}(i \in S_c)$:

$$\pi_{zi}^{\mathsf{STACK}} = \mathbb{P}(D_i = 1 \mid \mathbf{x}_i, i \in S) = \frac{\pi_{ci}}{\pi_{ci} + \pi_{ri}}$$

- The known reference probability π_{ri} acts as an **exact structural offset**, anchoring neural network logits $\eta_\theta(\mathbf{x}_i)$ to the true population scale.

The Five Sample Combination Formulations (Beresovsky et al., 2026)

Estimator	Sample Domain	Target Indicator	Composite Probability π_{zi}
`STACK`	$S_c \oplus S_r$ (Unlinked)	$I_{zi} = D_i$ (sample source)	$\frac{\pi_{ci}}{\pi_{ci}+\pi_{ri}}$
`LINK1`	$S_c \cup (S_r \setminus S_c)$	$I_{zi} = I_{ci}$ (deduplicated)	$\frac{\pi_{ci}}{\pi_{ci}+\pi_{ri}-\pi_{ci}\pi_{ri}}$
`LINK2`	$S_c \oplus S_r$ (Linked Stacked)	$I_{zi} = D_i + (1-D_i)I_{ci}$	$\frac{\pi_{ci}(1+\pi_{ri})}{\pi_{ci}+\pi_{ri}}$
`MIP`	Missing Information	$\tilde{I}_{zi}^{\text{MIP}} = D_i + (1-D_i)\tilde{p}_i$	Bridges `STACK` and `LINK2`
`PSEUDO`	Survey Weighted	Score: $\nabla \ell_{\text{pseudo}} = 0$	Design-weighted: $\sum_{S_r} \pi_{ri}^{-1}\text{softplus}(\eta_i)$

How MIP Bridges STACK and LINK2 vs. Why PSEUDO Explodes

Louis's (1982) Missing Information Principle (MIP):

- Imputes linkage expectations on unlinked reference units:

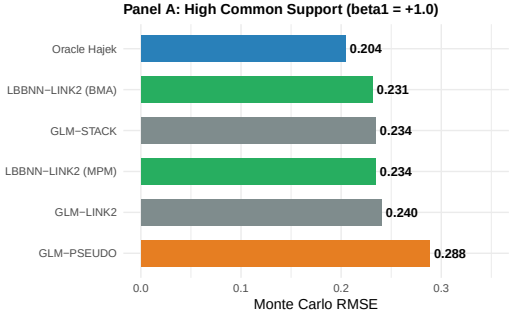
$$\tilde{p}_i = p_{\text{link}} I_{ci} + (1 - p_{\text{link}})\text{sg}(\pi_{ci}), \qquad \tilde{I}_{zi}^{\text{MIP}} = D_i + (1 - D_i)\tilde{p}_i$$

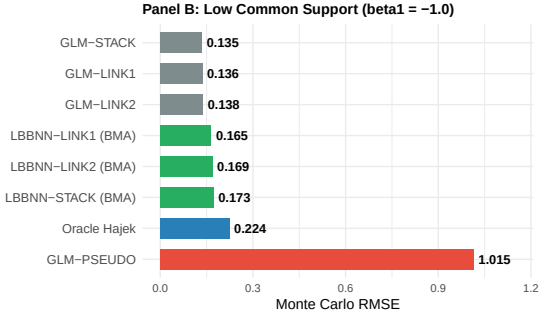
- Continuously bridges unlinked STACK ($p_{\text{link}} = 0$) and deterministically linked LINK2 ($p_{\text{link}} = 1$).

The Pseudo-Likelihood (PSEUDO) Breakdown:

- PSEUDO scores reference units with inverse design weights π_{ri}^{-1}.
- Under low common support (where $\pi_{ci} \gg 0$ but $\pi_{ri} \approx 0$), weights $\pi_{ri}^{-1} \to \infty$, causing loss gradients and parameter variance to explode.

Study 0 — Exact Linear Baseline Monte Carlo ($R = 20$, Visual Benchmark)





Empirical Findings:

- **Efficiency Hierarchy**: LINK2 $\succ$ LINK1 $\succ$ STACK $\succ$ PSEUDO.
- **Pseudo-Likelihood Breakdown**: Under low common support, `GLM-PSEUDO` explodes to RMSE $= 1.015$, while composite likelihoods and `LBBNN` remain bounded (RMSE ≈ 0.13–0.17).
- **LBBNN Agreement**: On linear truth, `LBBNN` matches parametric GLM efficiency (RMSE $= 0.231$ vs. 0.240).

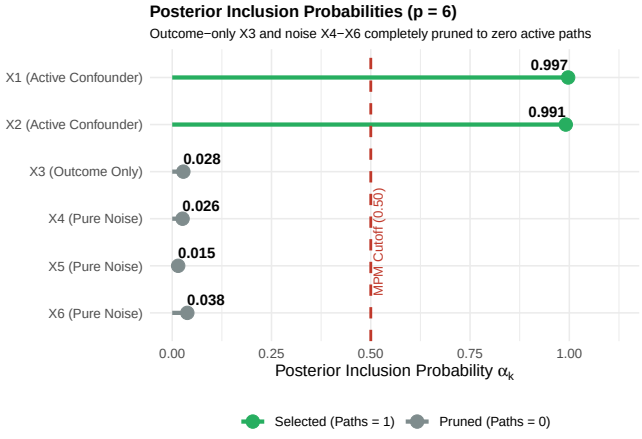
Study 1 — Automated Confounder Selection & Noise Pruning ($p = 6$)

Simulation Setup ($N = 5,000, p = 6$):

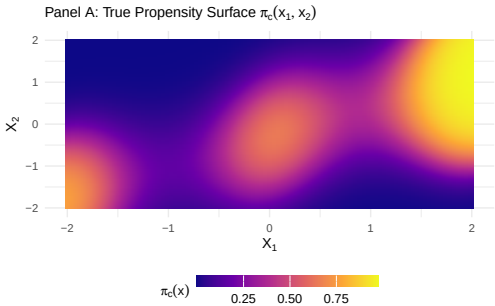
- **Confounders**: X_1, X_2 drive both propensity π_{ci} and outcome Y_i.
- **Outcome-Only**: X_3 drives Y_i but uncorrelated with selection I_c.
- **Pure Noise**: $X_4, X_5, X_6 \sim \mathcal{N}(0, 1)$.

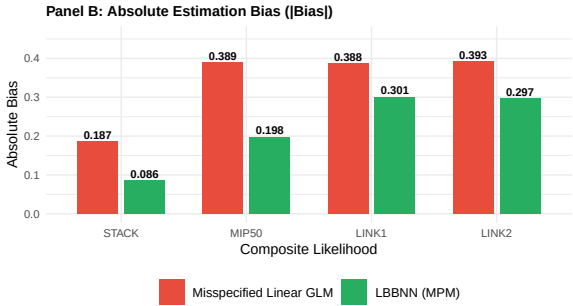
Why Variable Selection Matters:

- Including non-propensity predictors X_3–X_6 inflates Hájek weight variance without reducing confounding bias.
- `LBBNN` automatically prunes them.



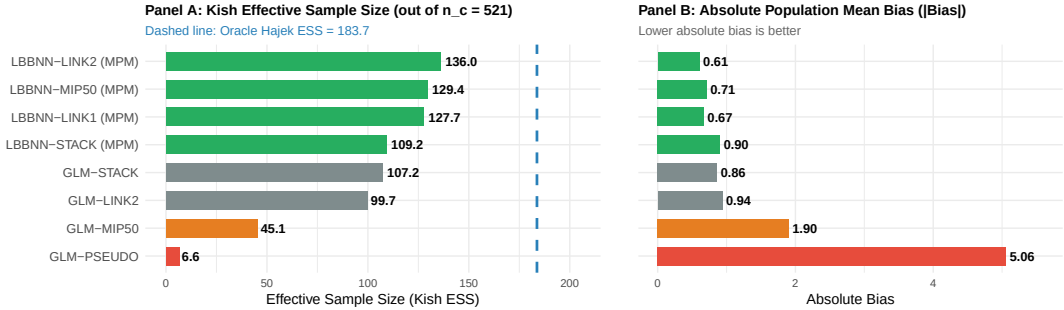
Study 2 — High Common Support with Non-Linear Propensity & Interaction





Non-Linear Recovery: Linear GLMs are structurally misspecified on the interaction surface ($|\text{Bias}| = 0.19$–0.39). `LBBNN` captures curvature and interaction, reducing bias across all composite likelihoods (**0.086** on `STACK`, 4.5x reduction!).

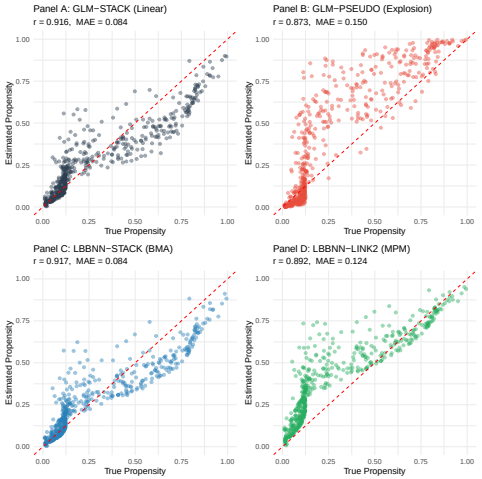
Study 3 — Low Common Support + Non-Linearity: Bias vs. Kish's ESS



Catastrophic Breakdown of Pseudo-Likelihood vs. Composite LBBNN:

- `GLM-PSEUDO` collapses Kish's ESS to **6.6 effective units** (out of $n_c = 521$), exploding bias to **+5.06**.
- `LBBNN-LINK2 (MPM)` achieves the **lowest bias (+0.61)** and **highest effective sample size (Kish ESS = 136.0)**.

Study 3 — Propensity Recovery Diagnostic (2x2 Scatterplot Grid)



Panels: (A) GLM-STACK bias from missing interactions; (B) GLM-PSEUDO severe weight explosion and scatter collapse; (C) LBBNN-STACK Bayesian Model Averaging calibration; (D) LBBNN-LINK2 Median Probability Model point recovery.

Section 4

PART IV: Unified Software Architecture & Synthesis

Unified LBBNN Architecture Across Track 1 & Track 2

Function / Module	Operational Track	Description & Statistical Role
`get_dataloaders()`	Track 1 & 2	Handles non-trainable weights $u_i \geq 0$, normalization, and multi-targets.
`lbbnn_net(sigma, ...)`	Track 1 & 2	ELBO likelihood scaling by error variance σ^2 (cold-posterior protection).
`loglik_contributions()`	Track 1B (Privacy)	Computes $S \times n$ record-level log-likelihood contributions $L_{s,i}$.
`fit_dp_lbbnn()`	Track 1B (Privacy)	Two-pass monotone surgical downweighting with Lipschitz bound Δ_α.
`sample_synthetic_lbbnn()`	Track 1B (Privacy)	Generates (ϵ, δ)-DP synthetic microdata via post-processing theorem.
`nll_link1()` / `nll_link2()`	Track 2 (Propensity)	Composite Bernoulli NLL on singly matched and matched-pair samples.
`make_nll_mip(p_link)`	Track 2 (Propensity)	Factory producing composite NLL under Missing Information Principle.
`nll_stack()` / `nll_pseudo()`	Track 2 (Propensity)	Stacked pooled sample NLL and survey design-weighted pseudo-NLL.
`build_propensity_dataset()`	Track 2 (Propensity)	Formats covariates and targets for any of the 5 estimation methods.
`hajek_estimate()`	Track 2 (Propensity)	Computes self-normalizing Hájek estimator $\sum w_i y_i / \sum w_i$ with $w_i = \hat{\pi}_{ci}^{-1}$.

Key Takeaways & Synthesis for Official Statistics

1. **Survey-Weighted Pseudo-Posteriors (Track 1A)**:
   - Normalizing survey weights to sample size ($\sum \tilde{w}_i = n$) + calibrated observation noise σ^2 prevents likelihood mis-scaling ($T \neq 1$) and protects spike-and-slab KL shrinkage.
   - `LBBNN` achieves 5x lower population RMSE than classical Hájek weighting under informative sampling while pruning irrelevant covariates.
2. **Two-Pass (ϵ, δ)-PDP Microdata Synthesis (Track 1B)**:
   - Monotone surgical downweighting (`fit_dp_lbbnn`) eliminates disclosure risk from extreme tail outliers ($\alpha_i = 0$) while preserving high utility ($\alpha_i \in [0.34, 0.85]$) on normal records.
3. **Composite Non-Probability Sample Integration (Track 2)**:
   - Conditioning on pooled sample inclusion $S_c \cup S_r$ allows known design probabilities π_{ri} to anchor neural network logits to the true population scale.
   - Composite Bernoulli likelihoods (`LINK2`, `LINK1`, `MIP`, `STACK`) bound loss gradients and prevent the catastrophic effective sample size collapse (`Kish ESS` < 7) of pseudo-likelihood under low common support.

References & Collaborative Works

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